Hadi Mohammadi

Senior AI & Data Science Expert · PhD in Explainable NLP

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Profile

Researcher in explainable NLP (PhD, Utrecht University) and Senior AI & Data Science Expert at AcademicTransfer. I build production LLM systems for ranking, matching, and evaluation, and I study how to make them transparent, reliable, and culturally fair.

Track record: six doctoral research projects, five competitively funded projects (over €40K) led end to end, and production AI used by 22 Dutch research universities and university medical centres. I release code and data with my work.

Professional Experience

AcademicTransfer, Senior AI & Data Science Expert - R&D Specialist — Utrecht, The Netherlands
Dec 2025 – present

- Lead the 2026 WBSO R&D programme as R&D Specialist (since Mar 2026), two projects approved for Mar–Dec 2026: an MCP-based academic query engine and an explainable CV matching engine with recruiters in the loop.
- Technical lead of the AI intelligence layer and applicant-tracking system used by 22 Dutch research universities and university medical centres.
- Own the AI stack: CV creation and revision, semantic search, talent-pool matching, and best-candidate identification.
- Lead a national recognition-and-rewards policy analysis with Dr. Sanli Faez (Universiteiten van Nederland) and university policy makers.
- Building an e-learning programme for PhD candidates and researchers with the University of Amsterdam.

AcademicTransfer, AI & Data Science Expert — Utrecht, The Netherlands
Jan 2024 – Dec 2025

- Cut HR hiring time by 20% with the CV Priority Sorter (active learning + LLMs), built with Utrecht University, TU Delft, and Maastricht University.
- Raised candidate–job matching accuracy by 17% and halved manual effort with generative pipelines for CV insight extraction and de-identification.
- Shipped transformer models on Azure ML with MLflow, Docker, and CI/CD; evaluated ranking with A/B tests, uplift analysis, and bandit simulations.

Utrecht University, School of Economics, Researcher & NLP Developer, FIRMBACKBONE — Utrecht, The Netherlands
Feb 2024 – Mar 2025

- Built the fbb-sustainability-analysis-cli tool: keyword extraction and sustainability scoring of company websites.

SnowaTec (Innovation Center), Entekhab Electronic, Senior Data Scientist, Customer Behaviour — Tehran, Iran
Dec 2021 – Jan 2023

- Modernised the enterprise data warehouse and ETL, improving data accessibility by 16%.
- Defined company KPIs and a customer-satisfaction index (+2% satisfaction); built Power BI dashboards used at every management level.
- Deployed ML customer segmentation, lifting marketing impact by 7.5%.

Bdood.bikes (shared bicycle system), Head of Data Science & Business Intelligence Team — Tehran, Iran
Apr 2020 – Oct 2020

- Led the data science and BI team and its analytics roadmap.
- Launched dynamic pricing and ML usage prediction, raising revenue by 5%.
- Cut maintenance costs by 8% and relocation costs by 3% with fleet intelligence; raised station utilisation by 11%.
- Improved customer retention by 4.5% through user-behaviour analyses.

Bdood.bikes (shared bicycle system), Data Scientist — Tehran, Iran
July 2019 – Apr 2020

- Reduced downtime by 3.8% with predictive maintenance and battery-health algorithms.
- Improved rider safety by 3% with GIS models identifying high-risk riders.
- Delivered marketing analytics behind a 6% rise in revenue and engagement.

Education

PhD **Utrecht University**, Explainable Natural Language Processing for Social Sciences — Utrecht, The Netherlands
Feb 2023 – Oct 2025

- Defended 9 September 2026.
- Thesis: *Let Me Explain! Explainable NLP for Understanding Large Language Models*.
- Department of Methodology & Statistics; member of the NLTP group and the Human Data Science Lab.
- Supervisors: Dr. Robert A. Bagheri, Prof. Dr. Daniel Oberski, Dr. Anastasia Giachanou.

MSc **University of Tehran**, Industrial Engineering, Macro Systems — Tehran, Iran
Sept 2018 – Apr 2021

- GPA 18.96/20 (4/4); ranked 2nd among 50+ graduate students.
- Thesis: dynamic pricing with demand learning via reinforcement learning under non-stationary customer behaviour.
- Cognitive science research on customer behaviour with the Institute for Cognitive Science Studies (ICSS).

BSc **Isfahan University of Technology, Golpayegan College of Engineering**, Mechanical Engineering — Isfahan, Iran
Sept 2011 – July 2016

Skills

Research: Explainable NLP, LLM evaluation and LLM-as-judge, fairness and cultural alignment, annotation reliability, human–AI collaboration.

ML & Statistics: Bayesian statistics, reinforcement learning, multi-armed bandits (Thompson sampling), transformers, topic modelling, A/B testing and uplift analysis, geospatial analysis, high-performance computing.

Engineering: Python (advanced), R, SQL, Bash; PyTorch, TensorFlow, Hugging Face, scikit-learn; RAG, agents, Model Context Protocol, vector databases; Azure ML, MLflow, Docker, CI/CD; Power BI; LaTeX.

Languages: Farsi (native), English (fluent), Dutch (A2, improving).

Publications

trajectory-judge: What Outcome-Only LLM Judges Miss on Agent Trajectories — 2026
Mohammadi, H.
arxiv.org/abs/2609.00038 (NeurIPS 2026 Workshop "Who Verifies the Agents?" (accepted))

Reasoning–Answer Entailment Under GRPO and DPO Fine-Tuning — 2026
Mohammadi, H., Giachanou, A.
mohammadi.cv/publications/grpo-dpo-cot (INLG 2026 (ACL), Utrecht, long paper, pp. 718–730)

EvalMORAAL: Interpretable Chain-of-Thought and LLM-as-Judge Evaluation for Moral Alignment in Large Language Models — 2026
Mohammadi, H., Giachanou, A., Bagheri, R. A.
mohammadi.cv/publications/evalmoraal (*SEM 2026 (ACL), pp. 497–515)

Exploring Cultural Variations in Moral Judgments with Large Language Models 2026
Mohammadi, H., Bagheri, R. A.
mohammadi.cv/publications/cultural-moral-judgments (Computational Linguistics in the Netherlands Journal, Vol. 15, pp. 59–77)

Assessing the Reliability of LLMs Annotations in the Context of Demographic Bias and Model Explanation 2025
Mohammadi, H.†, Shahedi, T.†, Mosteiro, P., Poesio, M., Bagheri, R. A., Giachanou, A.
mohammadi.cv/publications/llm-annotation-reliability (GeBNLP Workshop, ACL 2025, pp. 92–104 († equal contribution))

Do Large Language Models Understand Morality Across Cultures? 2025
Mohammadi, H., Meijer, Y. F. S. S., Papadopoulou, E., Bagheri, R. A.
mohammadi.cv/publications/morality-across-cultures (2nd LUHME Workshop, ECAI 2025, pp. 30–39)

Explainability-Based Token Replacement on LLM-Generated Text 2025
Mohammadi, H., Giachanou, A., Oberski, D. L., Bagheri, R. A.
mohammadi.cv/publications/ai-text-undetectability (arXiv preprint)

Explainability in Practice: A Survey of Explainable NLP Across Various Domains 2025
Mohammadi, H., Bagheri, R. A., Giachanou, A., Oberski, D. L.
mohammadi.cv/publications/explainability-survey (arXiv preprint; under review at the Journal of Information Science)

Measuring Firm Characteristics with Natural Language Processing (NLP) Models: An Error-Mitigating Framework with Tutorial Guidance 2025
Oberski, D. L., Domenech, J., Garcia-Bernardo, J., Gerbrands, P., Hassink, W., ***Mohammadi, H.***, Mustica, P., Schilpzand, R., van Witteloostuijn, A.
(Strategic Organization (under revision))

Novel Approaches in Financial Fraud Detection: Hybrid Machine Learning and Uncertainty-Based Deep Learning 2024
Mohammadi, H., Rahmati, M., Shahedi, T.
(BNAIC/BeNeLearn 2024, Utrecht, The Netherlands)

A Transparent Pipeline for Identifying Sexism in Social Media: Combining Explainability with Model Prediction 2024
Mohammadi, H., Giachanou, A., Bagheri, R. A.
mohammadi.cv/publications/sexism-detection (Applied Sciences, 14(19), 8620)

The CLIN33 Shared Task on the Detection of Text Generated by Large Language Models 2024
Fivez, P., Daelemans, W., Van de Cruys, T., Kashnitsky, Y., Chamezopoulos, S., ***Mohammadi, H.***, Giachanou, A., Bagheri, R. A., Poelman, W., Vladika, J., Ploeger, E., Bjerva, J., Matthes, F., van Halteren, H.
clinjournal.org/clinj/article/view/182 (Computational Linguistics in the Netherlands Journal, 13, 233–259)

Towards Robust Online Sexism Detection: A Multi-Model Approach with BERT, XLM-RoBERTa, and DistilBERT for EXIST 2023 Tasks 2023
Mohammadi, H., Giachanou, A., Bagheri, R. A.
ceur-ws.org/Vol-3497/paper-085.pdf (Working Notes of CLEF 2023, CEUR-WS Vol-3497, pp. 1000–1011)

Statistical Reinforcement Learning with Application in Dynamic Pricing Problem: Learning from Customer Interaction based on Bayesian Approach 2022
Mohammadi, H.
(Tehran: Arshadan Publication (book, in Persian))

Home Health Care Problem with Synchronization Visits and Considering Samples Transferring Time: A Case Study in Tehran, Iran 2022
Mirabnejad, M., ***Mohammadi, H.***, Mirzabaghi, M., Aghsami, A., Jolai, F., Yazdani, M.
10.3390/ijerph192215036 (International Journal of Environmental Research and Public Health, 19(22), 15036)

Grants & Awards

- **Best Oral Presentation**, IOPS Winter Conference 2025 — cultural moral alignment in LLMs (Dec 2025).
- **Applied Data Science Grant, €5K** — CV Priority Sorter for fair prioritization; with R. A. Bagheri, G. Krempl, J. Sparla (Dec 2025).
- **Applied Data Science Grant, €5K** — rule extraction from human evaluation with LLMs and RL; with A. Giachanou, S. Wang (Jun 2025).
- **CUCo Spark Grant, €9K** — ReDOSE: circular pharmaceutical industry using AI; with WUR, TU/e, and UMCU collaborators (Jul 2025).
- **ENFIELD grant, €14.4K** — European Lighthouse on Trustworthy and Green AI: user perspectives on explainable AI (Dec 2024).
- **Applied Data Science Grant, €7.5K** — reliability of annotations under model predictions and explanations; with P. Mosteiro Romero, A. Giachanou, M. Poesio (Dec 2023).
- **Admission scholarship**, Explainable AI Summer School, TU Delft (Aug 2022).
- **Ranked 2nd** among 50+ Industrial Engineering graduate students, University of Tehran (2021).
- **Best Paper Award**, 2nd National Conference of the Iranian System Dynamics Society (2019).
- **Ranked 35th** of 10,000+ in the national Iranian MSc entrance exam, Industrial Engineering (2018).

Teaching & Supervision

- **MSc thesis co-supervision, Applied Data Science, Utrecht University** — 2024: M. Meijer, E. Papadopoulou, L. Santos Moitinho de Almeida (with R. A. Bagheri); 2025: H. Brouwer, T. Kozák, M. Jouhar (with A. Giachanou); 2026: A. Vos, G. Verestiuc, T. Rietman (with R. A. Bagheri).
- **Practical Assistant**, Transformers Workshop, UMC Utrecht — Python sessions with Dr. Robert A. Bagheri and Prof. Dr. Albert Gatt (Dec 2023).
- **Computer Lab Assistant**, Data Wrangling and Data Analysis, MSc Applied Data Science, Utrecht University (Sep–Oct 2023).
- **Lead Teaching Assistant**, Deep Learning Summer School, Neuromatch Academy — international cohort, remote (Aug–Sep 2021).

Leadership & Service

- **Co-founder, NLP@U Young Scholar Network** — community of PhD candidates, postdocs, and junior NLP researchers across Utrecht University and UMC Utrecht (Nov 2024).
- **Communications Chair, BNAIC/BeNeLearn 2024** — Joint International Scientific Conferences on AI and Machine Learning, Utrecht University.
- **LERU Doctoral Summer School representative** for the Faculty of Social and Behavioural Sciences, Utrecht University (2025).

Professional Development

Generative Modeling Summer School, TU Eindhoven June 2024

Oxford Machine Learning Summer School (OxML 2023) — ML x Finance & NLP May – July 2023

Explainable AI Summer School, TU Delft — post-hoc interpretability August 2022

Executive Master in Finance, Sharif University of Technology Nov 2021 – Nov 2022

References

- Dr. Robert A. Bagheri — Associate Professor, Methodology & Statistics, Utrecht University — a.bagheri@uu.nl
- Dr. Anastasia Giachanou — Assistant Professor, Methodology & Statistics, Utrecht University — a.giachanou@uu.nl
- Prof. Dr. Daniel Oberski — Professor of Social Data Science, Utrecht University; Scientific Director, ODISSEI — d.l.oberski@uu.nl